

Chapter 5

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May 22, 2024

1 Seperable Spaces

Problem 5.1. Show that A is dense in X if and only if every nonempty open set of X contains a point of A

Proof. $\implies$: Suppose A is dense in X . Thus $\forall U \in \mathcal{T} \ni U \neq \emptyset, \forall p \in U$ p is a limit point of A or $p \in A$ thus $(U \setminus \{p\}) \cap A \neq \emptyset \implies U \cap A \neq \emptyset$. Thus $\exists a \in A \ni a \in U$.

$\impliedby$: Suppose $\forall U \in \mathcal{T}, \exists a \in A \ni a \in U \cap A$. $\forall x \in X \setminus A, \exists U \in \mathcal{T} \ni x \in U$, by the fact above $\exists a_x \in A \ni a_x \in U$, thus $a_x \in (U \setminus \{x\}) \cap A \neq \emptyset$. Thus x is a limit point of A . Thus $\overline{A} = X \implies A$ is dense in X . $\square$

Problem 5.2. Show $\mathbb{R}_{std}$ is separable. With which of the topologies on $\mathbb{R}$ that you have studied is $\mathbb{R}$ not separable?

Solution. $\forall x \in \mathbb{R} \setminus \mathbb{Q}$, it suffices to look at all sets $(x - \epsilon, x + \epsilon) \forall \epsilon > 0$. $\forall \epsilon > 0$, Let $a = x$ and $b = x + \epsilon$.

1. If $a > 0$: Thus $y - x > 0$. By the Archimedean property, $\exists n \in \mathbb{N} \ni n > \frac{1}{\epsilon} = \frac{1}{b-a} \implies nb - na > 1$. Thus $na + 1 < nb$. Since $a > 0, na > 1 \implies \exists m \in \mathbb{N} \ni m - 1 < na < m$. Thus $nb > na + 1 \geq m \implies nb > m > na \implies b > \frac{m}{n} > a$
2. If $a \leq 0$: $\exists k \in \mathbb{N} \ni k > |x|$. Thus $k - |x| = k + x > 0$ thus $k + x < k + y$. By the above $\exists r \in \mathbb{Q} \ni k + x < r < k + y$. Thus $\exists r' \in \mathbb{Q} \ni r' = r - k$. Thus $x < r' < y$.

Thus $\exists r \in \mathbb{Q} \ni r \in (a, b) \subseteq (a - \epsilon, a + \epsilon)$. Thus x is a limit point of $\mathbb{Q}$. Thus $\overline{\mathbb{Q}} = \mathbb{R} \implies \mathbb{Q}$ is dense in $\mathbb{R}$ with standard topology. $\mathbb{Q}$ is countable thus $\mathbb{R}$ is separable with standard topology.

1. Lower limit is also separable.
2. Discrete Topology: Not separable because all sets are open and closed. Thus the only set whose closure is $\mathbb{R}$ is $\mathbb{R}$ itself which isn't countable.
3. Finite Complement: Any infinite set is dense in $\mathbb{R}$ with this topology. Thus $\mathbb{R}$ is separable
4. Countable Complement: Any uncountable set is dense so take $[0, 1]$.

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Problem 5.4. Find a separable space that contains a subspace which isn't separable in the subspace topology.

Proof. Sorgenfrey Plane is separable but the negative diagonal is not because its subspace topology is Discrete. $\square$

Claim 5.5. If X and Y are separable spaces then $X \times Y$ is separable

Proof. Since X and Y are separable, $\exists M \subseteq X \ni |M| = |\mathbb{N}|, \overline{M} = X$ and $\exists N \subseteq Y \ni |N| = |\mathbb{N}|, \overline{N} = Y$. Thus, $M \times N$ is countable and $|M \times N| = |M| \times |N| = \aleph_0 \times \aleph_0 = \aleph_0$. Thus $X \times Y$ is separable. $\square$

TBD

Claim 5.6. The space $2^{\mathbb{R}}$ is separable

Proof. Let $A \subseteq 2^{\mathbb{R}}$ be the subset of functions which are fixed to 1 at all but a finite number of rational points or 0 at all but a finite number of rational points. I.e. $\forall f \in A, \exists B \subseteq \mathbb{Q}$, where B is finite such that $f(\mathbb{R} \setminus B) = 1$ or $f(\mathbb{R} \setminus B) = 0$. Let $A_n :=$ the set of functions which are fixed to 1 at all but n rational points and $B_n :=$ the set of functions which are 0 at all but n rational points. Thus $A = \bigcup_{n \in \mathbb{N}} (A_n \cup B_n)$.

Base Case: $|A_1| = 2 + |\mathbb{Q}| = |\mathbb{N}|$ and $|B_1| = 2 + |\mathbb{Q}| = |\mathbb{N}|$

Inductive Hypothesis: A_n and B_n are countable

Inductive Step: A_{n+1} is the set of functions which are 1 at all but $n + 1$ rational points so it union of countable sets of sets of A_1 and A_n . Thus A_{n+1} is countable. Similarly B_{n+1} is countable.

Thus A is the countable union of countable sets thus A is countable.

The open sets of $2^{\mathbb{R}}$ is given by the basis

$$\{f \in 2^{\mathbb{R}} \mid f(x_0) = a_1, \dots, f(x_n) = a_n, n \in \mathbb{N}, x_i \in \mathbb{R}, a_i \in \{0, 1\}\}$$

. Since open sets of $2^{\mathbb{R}}$ are given by infinite unions of the basis it suffices to prove that each basis intersects A nontrivially.

$\forall B \in \mathcal{B}, \exists U = \{u_1, \dots, u_n\} \subset \mathbb{R}$, a finite subset, and $(d_1, \dots, d_n) \in \{0, 1\}^n$, such that $\forall f \in B, f(u_i) = d_i$. Let $V \subseteq U$, be the set of u_i which are rational. $\square$

2 2^{nd} countable spaces

Claim 5.9. *Let X be 2^{nd} countable then X is separable*

Proof. Suppose X is 2^{nd} countable. Thus $\mathcal{B} = \{B_i\}_{i \in \mathbb{N}} \ni B_i \neq \emptyset$ for all $i \in \mathbb{N}$. Let $D = \{d_i\}_{i \in \mathbb{N}}$ where $d_i \in B_i, \forall i \in \mathbb{N}$. It suffices to prove that $\forall i \in \mathbb{N}, B_i \cap D \neq \emptyset$, since all open sets are unions of basis open sets. But $d_i \in B_i \cap D \neq \emptyset$ for all $i \in \mathbb{N}$. D is by definition countable. By 5.1 D is dense. Thus by definition of separable, X is separable. $\square$

Problem 5.10a. $\mathbb{R}$ is 2nd countable

Solution. By 3.2a, $\mathcal{B} = \{(a, b) \subset \mathbb{R} \mid a, b \in \mathbb{Q}\}$ is a basis. There exists a injective map $f : \mathcal{B} \rightarrow \mathbb{Q} \times \mathbb{Q}$ where $(a, b) \rightarrow \{a\} \times \{b\}$. Thus since $\mathbb{Q} \times \mathbb{Q}$ is countable, $\mathcal{B}$ is countable thus $\mathbb{R}$ is 2nd countable. $\blacksquare$

Problem 5.10b. $\mathbb{R}_L$ is not 2nd countable

Proof. Let $\mathcal{B}$ be a generic basis of $\mathbb{R}_L$. $\forall x \in \mathbb{R}, x \in [x, \infty) \in \mathcal{T}$, by the definition of basis $\exists B_x \in \mathcal{B} \ni x \in B_x \subseteq [x, \infty) \implies \min B_x = x$. Since $\forall x, y \in \mathbb{R}, x \neq y, \min B_x \neq \min B_y \implies B_x \neq B_y$. Thus $|\mathcal{B}| \geq |\mathbb{R}| > \aleph_0$. Thus $\mathbb{R}_L$ is not 2nd countable. $\square$

Problem 5.10c. H_{bub} is not 2nd countable

Proof. Note $B_{(x,y)}(1) = B(x, y)$ to simplify notation. Let $X := \{(x, 0) \mid x \in \mathbb{R}\}$. Let $\mathcal{B}$ be a generic basis for H_{bub} . $\forall x \in \mathbb{R}, (x, 0) \in B(x, 1) \subseteq \mathcal{T}$. By the definition of basis, $\exists B_x \in \mathcal{B} \ni (x, 0) \in B_x \subseteq B(x, 1)$ this implies $B_x \cap X = \{(x, 0)\}$.

$\forall x, y \in \mathbb{R} \ni x \neq y, B_x \cap X = \{(x, 0)\}$ and $B_y \cap X = \{(y, 0)\}$ Thus $B_x \neq B_y$.

Thus $|\mathcal{B}| \geq |\mathbb{R}| > \aleph_0$. Thus H_{bub} is not second countable. $\square$

Claim 5.11. *Every uncountable set in a 2nd countable space has a limit point*

Proof. Let X be the topological space which is 2nd countable. Let $A \subseteq X$ be a uncountable set. Thus X is uncountable. Let $\mathcal{B}$ be its countable basis. Assume the contrary that the theorem isn't true. Thus A has no limit points. Thus $\forall x \in A, \exists U_x \in \mathcal{T} \ni (U_x \setminus x) \cap A = \emptyset$. By definition of basis $\exists B_x \in \mathcal{B} \ni x \in B_x \subset U_x \implies B_x \cap A = \{x\}$. Let $C = \{B_x \mid x \in A\}$. C is countable because $C \subseteq \mathcal{B}$ and $\mathcal{B}$ is countable. Since $\forall x \in A, B_x \cap A = \{x\}$, $A = \bigcup_{B \in C} B \cap A$. The countable union of countable sets is countable, thus A is countable which is a contradiction. Thus $\exists x \in A$, a limit point of A . $\square$

3 1st countable spaces

Claim 5.14. *Let X be a 2nd countable space. Then X is 1st countable*

Proof. Suppose X is 2nd countable. Thus $\exists \mathcal{B}$, a basis for the topology of X , such that $\mathcal{B}$ is countable. Thus $\mathcal{B} = \{B_i\}_{i \in \mathbb{N}}$. $\forall x \in X$, $\mathcal{B}_x := \{B \in \mathcal{B} \mid x \in B\}$. $\mathcal{B}_x$ is countable since $\mathcal{B}$ is countable. $\forall B \in \mathcal{B}_x$, $x \in B$, by definition. $\forall U \in \mathcal{T} \ni x \in U$, by definition of basis, $\exists B \in \mathcal{B} \ni x \in B \subseteq U \implies B \in \mathcal{B}_x$, thus $\forall U \in \mathcal{T} \ni x \in U$, $\exists B \in \mathcal{B}_x \ni x \in B \subseteq U$. Thus $\mathcal{B}_x$ is a countable neighborhood basis for x . Thus X is 1st countable. $\square$

Claim 5.15. *If X is a topological space, $p \in X$, and p has a countable neighborhood basis, then p has a nested countable neighborhood basis.*

Proof. $\mathcal{B}_p := \{B_i\}_{i \in \mathbb{N}}$. Define $S_n := \bigcap_{i=1}^n B_i$. Thus Define $S := \{S_n\}_{n \in \mathbb{N}}$. Since $p \in B_i$ for all i , $p \in S_n$ for all n . Moreover $\forall U \in \mathcal{T} \ni p \in U$, $\exists B_i \in \mathcal{B}_p \ni B_i \subseteq U \implies S_i \subseteq B_i \subseteq U$. Thus S is a neighborhood basis. Moreover $S_n \supseteq S_{n+1} = S_n \cap B_{n+1}$. S_n is countable. Thus p has a nested countable neighborhood set. $\square$

Problem 5.16a. $\mathbb{R}_L$ is 1st countable

Proof. $\forall x \in \mathbb{R}$. $\mathcal{B}_x := \{[x, x + \frac{1}{n})\}_{n \in \mathbb{N}}$. $\mathcal{B}_x$ is countable. $\forall n \in \mathbb{N}$, $x \in [x, x + \frac{1}{n})$. $\forall U \in \mathcal{T} \ni x \in U$, by definition of basis $\exists \epsilon > 0 \ni [x, x + \epsilon) \subseteq U$. By the Archimedean property, $\exists K \in \mathbb{N} \ni K > \frac{1}{\epsilon}$. Thus $[x, x + \frac{1}{K}) \subseteq [x, x + \epsilon) \subseteq U$. Thus $\mathcal{B}_x$ is a countable neighborhood basis. Thus $\mathbb{R}_L$ is 1st countable. $\square$

Claim 5.18. *Suppose x is a limit point of the set A in a 1st countable space X . Then there is a sequence of points $\{a_i\}_{i \in \mathbb{N}} \subseteq A$ that converges to x .*

Proof. Since X is 1st countable, $\exists \mathcal{B}_x$ a countable neighborhood basis for x . WLOG (claim 5.15), $\mathcal{B}_x$ is nested. Since x is a limit point of A , $\forall i \in \mathbb{N}$, $\exists a_i \in (B_i - \{x\}) \cap A$. Thus make a sequence with these a_i . $\{a_i\}_{i \in \mathbb{N}}$ is by definition in A . Furthermore by the definition of a nested countable neighborhood basis $\forall U \in \mathcal{T}$, $\exists K \in \mathbb{N} \ni B_K \subseteq U$. Since $n \geq K \implies B_n \subseteq B_K \implies a_n \in B_n \subseteq U$. Thus $\forall n \geq K$, $a_n \in U$. Thus $\{a_i\}$ converges to x . $\square$

4 Souslin Property

Problem 5.21. *Show that $\mathbb{R}$ is Souslin*

Proof. Assume the contrary that $\mathbb{R}$ is not Souslin. Thus $\exists C \subseteq \mathcal{T} \ni \forall U, V \in C \ni U \neq \emptyset \neq V, U \neq V \implies U \cap V = \emptyset$ and C is uncountable. Since $\mathbb{R}$ is separable $\exists D$ a dense countable set. Since $U \cap V = \emptyset \implies (U \cap D) \cap (V \cap D) = \emptyset$. Create the set E using the following rule, $\forall U \in C$, pick a element of $U \cap D$. Each element of C will give a unique element. Thus $E \subseteq D$ but $|C| = |E| > \aleph_0$, which is a contradiction since D is countable. Thus $\mathbb{R}$ must be Souslin. $\square$

Claim 5.22. *A separable space has the Souslin property.*

Proof. Same proof as above but without using $\mathbb{R}$ as set. $\square$